

ACME Corporation – Cloud Migration Plan Document

Strategic Objective

The purpose of this cloud migration project is to migrate ACME Corporation's internal cybersecurity laboratory infrastructure to the Microsoft Azure cloud platform in a secure and efficient manner. The migration has been proposed to focus in two phases, beginning with a "bare metal" deployment of three virtualized assets – Kali Linux, Metasploitable 2, and Windows Server 2022 - as a guide to establish a hardened baseline environment for the purpose of cyber education, penetration testing and incident response simulation. The migration will focus on a cloud security first principle that encompasses operational scale, fault tolerance, and governance compliance while ensuring minimal interruption of ACME Corporation's existing infrastructure.

Worked with Eric Solorzano but we both had issues with the uploading part of the VM.

Kali Linux Hardening Summary

- **SSH Key-Only Authentication:**
 - Generated id_ed25519 key pair
 - Public key installed in ~/.ssh/authorized_keys
 - SSH directory (~/.ssh) set to chmod 700; authorized_keys to chmod 600
- **SSH Daemon Configuration (/etc/ssh/sshd_config):**
 - PubkeyAuthentication yes
 - AuthenticationMethods publickey
 - ChallengeResponseAuthentication no
 - PermitEmptyPasswords no
- **SSH Login Test:**
 - Verified remote access with keys only
 - Updated known_hosts with server fingerprint
- **Fail2Ban Installation:**
 - Installed and configured fail2ban to mitigate brute-force attacks
 - Set bantime = 10m, findtime = 10m, maxretry = 5
- **Local Password Hardening:**

- Set complex password: M0on\$h!n3_T!ger7
- Length: 14 characters with upper/lowercase, symbols, and numbers
- **Snapshot Taken:**
 - VM snapshot saved post-configuration

Metasploitable 2 Hardening Summary

- **Service Removal:**
 - Stopped and removed vulnerable services using update-rc.d and killall:
 - vsftpd, telnetd, apache2, smbd, postgresql, mysql, tomcat, vncserver, nfs-kernel-server, rpcbind
 - Verified shutdown with netstat -tulnp after each removal
- **SSH Configuration:**
 - Updated /etc/ssh/sshd_config:
 - PermitRootLogin no
 - PasswordAuthentication no (optional/planned)
 - PubkeyAuthentication yes
 - Restarted SSH service with sudo service ssh restart
- **Password Update:**
 - Replaced default msfadmin credentials with complex password: R3d\$P@nth3r!92
- **Firewall Installation (UFW):**
 - Default deny inbound, allow outbound
 - Allowed only port 22 (SSH)
 - Enabled and validated rules persist across reboot
- **Snapshot Taken:**
 - Post-hardening image captured

Windows Server 2022 Hardening Summary

- **Initial Configuration:**
 - Set strong admin password
 - Renamed system and configured local static IP
 - Installed all pending system updates

- **Domain Controller Promotion:**
 - Installed Active Directory Domain Services (AD DS) and DNS
 - Promoted server to domain controller for acmecollege.com
- **User & Credential Security:**
 - Created 5 domain users with strong passwords (14+ chars, symbols)
 - Created backup admin with password never expires
 - Added users to a custom security group
- **Group Policy Object (GPO) Configuration:**
 - **Password Policy:**
 - Minimum length: 14
 - Complexity: Enabled
 - Max age: 90 days
 - Password history: 24
 - **Account Lockout:**
 - Lockout threshold: 5
 - Lockout duration: 15 min
 - Reset counter: 10 min
 - **Audit Policy:**
 - Enabled audit for logon, privilege use, object access, and policy changes
 - **User Rights Assignment:**
 - Denied Guest login
 - Allowed RDP for Admins only
 - **Security Options:**
 - Disabled guest account
 - Enforced Ctrl+Alt+Del
 - Blocked anonymous SAM enumeration
- **RDP Hardening:**
 - Enabled RDP
 - Configured IP scope in firewall to limit RDP access
 - Enabled NLA (optional toggle for compatibility)
- **Firewall Configuration:**
 - Allowed: TCP 3389 (RDP), TCP/UDP 53 (DNS)

- Disabled SMB (445), FTP (21), and other legacy services
- Used Advanced Firewall to apply GPO rules
- **Disabled Unused Services:**
 - Print Spooler, Bluetooth Support, Fax, Remote Registry, WSD Print, and Windows Remote Management (if unused)
- **Logging & Snapshot:**
 - Snapshot created post-hardening

3. Migration Toolchain & Conversion Workflow

The migration leveraged a combination of open-source and cloud-native tools to convert, upload, and deploy the virtual machines:

- **VM Format Conversion**
 - Used StarWind V2V Converter to transform .vmdk disks into fixed-size .vhd format suitable for Azure Hyper-V-based deployments.
 - Selected pre-allocated (fixed-size) disks to avoid runtime expansion issues during cloud deployment.
- **Azure Upload & Disk Storage Configuration**
 - Configured Azure Storage Account with secure transfer enabled
 - Used Azure Storage Explorer to upload VHDs as a disk
- **Future-Ready Automation**
 - Azure Migrate was scoped for enabling seamless discovery, dependency mapping, and automated migration pipelines for large-scale replication of remaining VMs.

Risk Mitigation & Compliance Controls

To minimize the operational, security, and compliance risks during migration, the following safeguards were implemented:

- **Snapshot Rollback Strategy**

- Each system snapshot was saved post-hardening and pre-migration to serve as a rollback checkpoint in case of misconfiguration or corruption during transfer.
- **Compliance Alignment**
 - System configurations were designed with GDPR, FERPA, and ISO/IEC 27001 control mappings in mind, including encryption, account lockouts, and least-privilege access enforcement.
- **Vulnerability Audits Pre-Deployment**
 - Local scanning tools (Nmap, netstat, and UFW logs) were used to validate the attack surface before enabling cloud access.

Migration Timeline & Scaling Strategy

This migration is structured in two major phases, with clearly defined timelines:

Phase	Task Scope	Timeframe
Initial Batch	Harden, convert, upload, and deploy first 3 VMs	April 20 – May 4
Ongoing Migration	Migrate 3 additional VMs weekly (same workflow)	May 5 – June 30

- The target is to migrate and deploy 20 fully hardened VMs by the end of June, establishing a comprehensive and scalable cloud-based security lab.
- Each upload cycle includes:
 - Security hardening & snapshot
 - V2V conversion
 - Azure blob upload
 - Deployment and verification
 - Documentation update